Supplementary material

Mesh-to-Mesh Transfer Errors in Multistep Unsteady Flow Simulations

This supplement contains the full Euler remeshing comparisons and additional tests of the transferred history states supporting the article. The tables use the archived numerical data supplied with the article. NN, BAR, GP, and CV denote nearest-neighbor Taylor interpolation, barycentric interpolation, Galerkin projection, and control-volume transfer; suffixes give nominal order of accuracy for smooth fields. GC denotes bounded global correction; LC denotes locally conservative flux correction. BDF2 denotes the second-order backward differentiation formula.

S1 Repeated remeshing in Euler simulations

Table S1 includes the higher-order alternatives to the main-text comparison, and Table S2 reports all three realizations, including globally corrected NN1. The remeshing interval is given in physical time steps. The deformation parameter s bounds streamwise displacement by the smallest incident corner-triangle altitude divided by s; $s = 2$ permits twice the displacement of $s = 4$. The wide-window position is the 50% crossing, and apparent width is the distance between the 10% and 90% crossings using plateaus in $W_1 = [2.40, 3.00]$. At 398 transfers, contact smearing contaminates the left plateau for NN1 and BAR2, so these widths are not shock-only measures. Differences from the fixed-mesh calculation are expressed in cells of spacing $h = 0.01$. With $Q_M(\rho) = \sum_i w_i^M \rho_i$, where w_i^M is the median-dual volume, $\sum \Delta I_\rho / I_{\rho,0}$ is the signed sum of target-minus-source density-integral changes immediately across transfer, normalized by initial mass and excluding physical evolution. Integral cancellation leaves local distortion unconstrained, so the tables report both integral and profile errors.

Table S1: Frequency comparison at safety factor $s = 4$ for all transfer variants. Δn is the number of physical time steps between mesh changes; $\Delta x/h$ and $\Delta w/h$ are signed wide-window position and apparent-width changes relative to the fixed-mesh reference. Widths can include contact smearing and must not be read as shock thickness.

Transfer	Δn	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho / I_{\rho,0}$
NN1	10	0.14	0.57	$1.1 \cdot 10^{-4}$
NN1	2	0.1	21.41	$5 \cdot 10^{-4}$
NN2	10	−0.02	−0.01	$-8.8 \cdot 10^{-6}$
NN2	2	−0.04	0	$-1.7 \cdot 10^{-5}$
NN3	10	0.01	−0.01	$2.2 \cdot 10^{-5}$
NN3	2	0.02	0	$2.7 \cdot 10^{-5}$
BAR2	10	0.01	0.25	$2 \cdot 10^{-5}$
BAR2	2	−0.03	3.24	$1.4 \cdot 10^{-4}$
BAR3	10	0	0.01	$-1.7 \cdot 10^{-5}$
BAR3	2	0	0.07	$-5.2 \cdot 10^{-6}$
GP2	10	0.01	0	$2.6 \cdot 10^{-6}$
GP2	2	0.01	0	$-2.8 \cdot 10^{-6}$
GP3	10	−0.01	0	$-2.1 \cdot 10^{-5}$
GP3	2	−0.01	0.02	$1.8 \cdot 10^{-5}$
CV2	10	0	0	$3 \cdot 10^{-14}$
CV2	2	0	0.01	$-8.2 \cdot 10^{-15}$

Table S2: High-frequency Euler comparison across three random deformation realizations. GC indicates whether the global correction is applied; $s = 2$ permits twice the displacement amplitude of $s = 4$. Position and width changes are in cells of the $h = 0.01$ reference mesh. Widths use W_1 and include contact smearing for strongly diffused cases. The final column is the signed sum of transfer mass changes divided by initial mass; it excludes all CFD-step changes.

Transfer	GC	s	Realization	$\Delta x/h$	$\Delta w/h$	$\sum \Delta I_\rho / I_{\rho,0}$
BAR2	No	2	1	-0.06	22.91	$2.7 \cdot 10^{-4}$
BAR2	No	2	2	-0.03	22.98	$3.1 \cdot 10^{-4}$
BAR2	No	2	3	-0.07	22.9	$1.9 \cdot 10^{-4}$
BAR2	No	4	1	-0.03	3.24	$1.4 \cdot 10^{-4}$
BAR2	No	4	2	0	3.2	$1.7 \cdot 10^{-4}$
BAR2	No	4	3	-0.04	3.12	$6.6 \cdot 10^{-5}$
CV2	No	2	1	0.19	0.06	$-9.5 \cdot 10^{-15}$
CV2	No	2	2	0.2	0.07	$-4.2 \cdot 10^{-14}$
CV2	No	2	3	0.18	0.05	$7.9 \cdot 10^{-14}$
CV2	No	4	1	0	0.01	$-8.2 \cdot 10^{-15}$
CV2	No	4	2	0.01	-0.01	$6.8 \cdot 10^{-14}$
CV2	No	4	3	0	0.01	$4.6 \cdot 10^{-14}$
GP2	No	2	1	0.17	0.06	$-1 \cdot 10^{-5}$
GP2	No	2	2	0.19	0.06	$-9.6 \cdot 10^{-6}$
GP2	No	2	3	0.19	0.03	$-9.5 \cdot 10^{-6}$
GP2	No	4	1	0.01	0	$-2.8 \cdot 10^{-6}$
GP2	No	4	2	0.02	0	$-3.1 \cdot 10^{-6}$
GP2	No	4	3	0.01	0.01	$-2.5 \cdot 10^{-6}$
NN1	Yes	2	1	-0.91	25.65	$-2.8 \cdot 10^{-15}$
NN1	Yes	2	2	-0.65	23.8	$7.9 \cdot 10^{-14}$
NN1	Yes	2	3	0.01	26.33	$1 \cdot 10^{-13}$
NN1	Yes	4	1	-0.15	18.43	$-5.3 \cdot 10^{-14}$
NN1	Yes	4	2	-0.01	21.28	$9.5 \cdot 10^{-15}$
NN1	Yes	4	3	-0.11	21.2	$-1.9 \cdot 10^{-14}$
NN1	No	2	1	-0.38	26.36	$1.3 \cdot 10^{-3}$
NN1	No	2	2	-0.29	24.98	$-1.6 \cdot 10^{-3}$
NN1	No	2	3	0.44	27.14	$4.9 \cdot 10^{-4}$
NN1	No	4	1	0.1	21.41	$5 \cdot 10^{-4}$
NN1	No	4	2	0.3	22.89	$-8.9 \cdot 10^{-4}$
NN1	No	4	3	0.2	22.75	$8.3 \cdot 10^{-4}$

S2 Boundary residuals and the cancellation test

S2.1 Slip-wall comparison

At a common trial state, the predicted residual perturbation is $-M_B\eta^n/\Delta t$, with $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2$ formed from the errors in the latest and older states relative to the target-mesh reference. Here M_B is the target finite-volume mass matrix and Δt the physical time step. The solver reports the negative of this residual. The following comparisons measure the difference between the predicted and computed residual perturbations.

At the 370 slip-wall vertices, with both residuals evaluated at the same trial state, the wall-normal momentum mismatch is 3.4×10^{-12} for GP2, 4.1×10^{-12} for GP3 and CV2, and 4.5×10^{-12} for NN1. The tangential mismatch is $(1.4\text{–}1.7)\times10^{-11}$, a fraction $(1.1\text{–}1.3)\times10^{-7}$ of the tangential residual perturbation of 1.25×10^{-4}. All four transfers return the same wall state, that of the source solution; the difference in wall-normal momentum from the target-mesh reference is 1.1×10^{-14}, so the normal residual perturbation is negligible and enforcement of the target slip-wall condition does not alter it. The predicted and measured perturbations both include enforcement of the target slip-wall condition.

S2.2 Residual probes in the cancellation experiment

In the cancellation experiment, the history discrepancies are $\varepsilon^n = e/4$ and $\varepsilon^{n-1} = e$, giving $\eta^n = 2\varepsilon^n - \varepsilon^{n-1}/2 = 0$. The initial residual probes nevertheless use different trial states. Let M_B be the target finite-volume mass matrix and F_B its spatial residual. For a trial state Y and displacement q, their difference has the form

$$\Delta\Phi = \frac{3M_Bq}{2\Delta t} + F_B(Y+q) - F_B(Y) - \frac{M_B\eta^n}{\Delta t}. \tag{S1}$$

With $q = e/4$ and $\eta^n = 0$, the temporal part is three quarters of the magnitude of the residual perturbation with only the older state in error. The measured ratio near 0.75 also includes a spatial-residual difference and therefore does not verify the residual identity at a common trial state. In contrast, at an identical trial state the history contribution to the residual difference is exactly $-M_B\eta^n/\Delta t$. Cancellation is temporary even for an exact first step: if the first-step state difference vanishes, the older level retains $e/4$, producing the subsequent history combination $-e/8$.

S3 Perturbation-amplitude dependence

Let δ denote the state difference from the target-mesh reference. Subscripts f, n, and $n-1$ denote equal spatial discrepancies imposed in both BDF2 history states, the latest only, and the older only. The superposition remainder is $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$; its lift counterpart is $\mathcal{E}_L = \Delta C_{L,f} - \Delta C_{L,n} - \Delta C_{L,n-1}$. State norms are Euclidean or volume-weighted, $\|v\|_M = (v^T M_B v)^{1/2}$. Scaling the discrepancy by $\lambda = 1, 1/2, 1/4$ reduces the superposition remainders in the Euclidean norm and in lift for BAR2, GP2, and CV2 (Table S3). The absolute remainder in the Euclidean norm decreases by factors of about 3.2–3.5 per halving, faster than linearly over this range. The absolute remainder in the volume-weighted norm falls more slowly, from approximately 2.4×10^{-5} to $1.5\text{–}1.7 \times 10^{-5}$, while its normalized value rises from about 3.4% to 8.7–9.2%. The departure from linear response depends on the quantity and norm examined. These tests do not establish a quadratic remainder in every norm, and no independently measured iterative error explains the remaining discrepancy.

Table S3: Amplitude dependence of first-step superposition. $\mathcal{E} = \delta_f - \delta_n - \delta_{n-1}$ is the state remainder and $\mathcal{E}_L$ its lift counterpart. Relative remainders use the corresponding response with both states perturbed. The Euclidean and volume-weighted state metrics have different amplitude dependence.

Transfer	λ	$\|\mathcal{E}\|_2$	$\|\mathcal{E}\|_2/\|\delta_f\|_2$	$\|\mathcal{E}\|_M$	$\|\mathcal{E}\|_M/\|\delta_f\|_M$	$\|\mathcal{E}_L\|$
BAR2	1	$1.76 \cdot 10^{-3}$	$7.78 \cdot 10^{-3}$	$2.36 \cdot 10^{-5}$	$3.36 \cdot 10^{-2}$	$2.22 \cdot 10^{-5}$
BAR2	0.5	$5.45 \cdot 10^{-4}$	$4.82 \cdot 10^{-3}$	$1.97 \cdot 10^{-5}$	$5.6 \cdot 10^{-2}$	$4.74 \cdot 10^{-6}$
BAR2	0.25	$1.6 \cdot 10^{-4}$	$2.83 \cdot 10^{-3}$	$1.54 \cdot 10^{-5}$	$8.74 \cdot 10^{-2}$	$1.76 \cdot 10^{-6}$
GP2	1	$1.86 \cdot 10^{-3}$	$8.23 \cdot 10^{-3}$	$2.45 \cdot 10^{-5}$	$3.44 \cdot 10^{-2}$	$2.23 \cdot 10^{-5}$
GP2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.42 \cdot 10^{-6}$
GP2	0.25	$1.65 \cdot 10^{-4}$	$2.93 \cdot 10^{-3}$	$1.65 \cdot 10^{-5}$	$9.24 \cdot 10^{-2}$	$1.58 \cdot 10^{-6}$
CV2	1	$1.86 \cdot 10^{-3}$	$8.22 \cdot 10^{-3}$	$2.42 \cdot 10^{-5}$	$3.4 \cdot 10^{-2}$	$2.27 \cdot 10^{-5}$
CV2	0.5	$5.82 \cdot 10^{-4}$	$5.15 \cdot 10^{-3}$	$2.06 \cdot 10^{-5}$	$5.77 \cdot 10^{-2}$	$4.53 \cdot 10^{-6}$
CV2	0.25	$1.66 \cdot 10^{-4}$	$2.95 \cdot 10^{-3}$	$1.64 \cdot 10^{-5}$	$9.16 \cdot 10^{-2}$	$1.63 \cdot 10^{-6}$

S4 RANS iterative convergence for each method

The Spalart–Allmaras (SA) and shear-stress-transport (SST) calculations use BDF2 equations solved by pseudo-time iteration at each physical time step. The stopping target is $\log_{10} \mathrm{rms}(r_\rho) \leq -12$ in the solver density-residual normalization. Reaching the iteration limit alone does not establish convergence. Table S4 gives the full records behind the compact main-article comparison; final residuals are medians over physical time steps, not residuals of averaged solutions. The nominal rows cover the full remeshing interval and other rows the shorter sensitivity interval; Δt_0 denotes the nominal physical time step.

Table S4: Iterative convergence by method for the first deformation realization. Residuals are median final $\log_{10} \mathrm{rms}(r_\rho)$; $N_{\max}$ is the iteration limit.

Closure	Transfer	$N_{\max}$	$\Delta t/\Delta t_0$	Physical steps	Steps at limit	Residual
SA	GP2	800	1	398	152	-12.00
SA	NN2	800	1	398	155	-12.00
SA	CV2	800	1	398	147	-12.00
SA	GP2	1600	1	38	0	-12.00
SA	NN2	1600	1	38	0	-12.00
SA	CV2	1600	1	38	0	-12.00
SA	GP2	800	1/2	78	0	-12.00
SA	NN2	800	1/2	78	0	-12.00
SA	CV2	800	1/2	78	0	-12.00
SA	GP2	800	1/4	158	0	-12.00
SA	NN2	800	1/4	158	0	-12.00
SA	CV2	800	1/4	158	0	-12.00
SST	GP2	800	1	398	333	-11.10
SST	NN2	800	1	398	324	-11.24
SST	CV2	800	1	398	336	-11.11
SST	GP2	1600	1	38	28	-11.33
SST	NN2	1600	1	38	28	-11.33
SST	CV2	1600	1	38	30	-11.27
SST	GP2	6400	1	38	28	-11.27
SST	NN2	6400	1	38	30	-11.17
SST	CV2	6400	1	38	29	-11.18
SST	GP2	800	1/2	78	56	-11.27
SST	NN2	800	1/2	78	58	-11.32
SST	CV2	800	1/2	78	60	-11.23

S5 Flux correction in repeated Euler transfers

Table S5 reports all 36 matched comparisons. Standard and LC cases use the same mesh trajectories. E_1 is the volume-weighted mean absolute density error against the exact Sod solution at step 399, and its paired percentage change is $100(E_{1,\mathrm{LC}}/E_{1,0} - 1)$. Position differences use the wide-window 50% crossing; widths in this table instead use the shock-only interval $[2.55, 2.85]$. Both are measured relative to the corresponding fixed-mesh reference in cells of spacing 0.01. The shock-only diagnostic uses the stored eight-line mean profile at 800 streamwise probes. At this time, the exact contact and shock are at approximately 2.370 and 2.699. For heavily smeared NN1/BAR2 profiles, contact diffusion can raise the wide-window left plateau from approximately 0.266 to 0.287 and shift the 10% crossing into the contact tail. A large wide-window width therefore does not imply equally large shock broadening.

Table S5: Matched standard and LC transfers. Δn is the remeshing interval, s the deformation parameter, and r the realization. Subscript 0 denotes the standard treatment. ΔE_1 is a paired percentage change; positions and shock-only widths are in reference cells.

Transfer	Δn	s	r	ΔE_1 (%)	$\Delta x_0/h$	$\Delta x_{\mathrm{LC}}/h$	$\Delta w_0/h$	$\Delta w_{\mathrm{LC}}/h$
BAR2	10	4	1	−0.14	0.01	0.01	0.14	0.14
BAR3	10	4	1	0	0	0	0	0
CV2	10	4	1	−0.51	0	−0.01	−0.02	−0.02
GP2	10	4	1	−0.73	0.01	−0.01	−0.01	−0.01
GP3	10	4	1	−0.35	−0.01	−0.01	−0.03	−0.03
NN1	10	4	1	−0.37	0.14	0.09	0.36	0.36
NN2	10	4	1	0.2	−0.02	−0.01	−0.03	−0.03
NN3	10	4	1	−0.16	0.01	0	−0.03	−0.02
BAR2	2	4	1	−0.4	−0.03	−0.02	1.14	1.13
BAR2	2	4	2	−0.29	0	−0.01	1.02	1.02
BAR2	2	4	3	−0.02	−0.04	−0.03	1.05	1.05
BAR3	2	4	1	−0.06	0	0.02	0.08	0.07
CV2	2	4	1	−1.23	0	−0.01	−0.01	−0.01
CV2	2	4	2	−1.31	0.01	0	0.01	0.01
CV2	2	4	3	−1.33	0	−0.01	−0.01	−0.01
GP2	2	4	1	−1.56	0.01	−0.01	−0.01	−0.01
GP2	2	4	2	−2.01	0.02	0	0	0.01
GP2	2	4	3	−1.93	0.01	0	−0.01	0
GP3	2	4	1	−0.99	−0.01	0	0.02	0.01
NN1	2	4	1	0.37	0.1	0.06	1.49	1.53
NN1	2	4	2	0.6	0.3	0.2	1.6	1.68
NN1	2	4	3	−1.82	0.2	0.11	1.19	1.17
NN2	2	4	1	1.34	−0.04	−0.03	−0.02	−0.03
NN3	2	4	1	0.45	0.02	0	−0.01	−0.01
BAR2	2	2	1	−0.62	−0.06	−0.06	2.28	2.28
BAR2	2	2	2	−0.55	−0.03	−0.05	2.17	2.16
BAR2	2	2	3	−0.38	−0.07	−0.07	2.17	2.17
CV2	2	2	1	−5.14	0.19	0.01	0.01	0.01
CV2	2	2	2	−5.41	0.2	0.01	0.03	0.05
CV2	2	2	3	−5.34	0.18	0	−0.01	0.03
GP2	2	2	1	−7.01	0.17	0.01	0.02	0.01
GP2	2	2	2	−7.86	0.19	0.02	0.03	0.04
GP2	2	2	3	−7.51	0.19	0.01	−0.02	0.03
NN1	2	2	1	−0.3	−0.38	−0.53	8.68	8.68
NN1	2	2	2	1.9	−0.29	−0.34	8.37	8.34
NN1	2	2	3	0.35	0.44	0.29	8	8.03

The direct final drifts are normalized by the exact mass 2.375 and energy 5.75. Across all matched LC cases their absolute values are below 5×10^{-13} and 7×10^{-13}, respectively, in the double-precision restart after the final transfer. This is distinct from summing instantaneous transfer defects. Table S6 separates the final CFD-step and transfer changes for representative cases. A BDF2 calculation also retains a difference between its two history levels; the complete archived budget includes that quantity. The table entries describe stage changes and final drift, not an additive sum of the entire multistep history.

Table S6: Relative mass changes in the final CFD step and transfer, and final drift, for realization 1 with remeshing every two steps. Standard is the usual transfer treatment; LC is the locally conservative flux correction. All values are normalized by initial mass.

Transfer	s	Treatment	CFD step	Transfer	Final drift
NN1	4	Standard	$-3.5 \cdot 10^{-7}$	$-1.4 \cdot 10^{-5}$	$1.4 \cdot 10^{-4}$
NN1	4	LC	$1.7 \cdot 10^{-15}$	$1.9 \cdot 10^{-16}$	$2.6 \cdot 10^{-13}$
GP2	2	Standard	$1.1 \cdot 10^{-8}$	$4.6 \cdot 10^{-7}$	$-3.9 \cdot 10^{-6}$
GP2	2	LC	$1.7 \cdot 10^{-15}$	$-1.9 \cdot 10^{-16}$	$4.2 \cdot 10^{-13}$
CV2	2	Standard	$2.7 \cdot 10^{-8}$	$0 \cdot 10^{0}$	$5.5 \cdot 10^{-6}$
CV2	2	LC	$0 \cdot 10^{0}$	$0 \cdot 10^{0}$	$-8 \cdot 10^{-15}$

The density extrema permitted by LC are measured relative to the source range at each transfer. NN1 and BAR2 lose strict monotonicity, while excursions remain within the 2×10^{-3} bound relaxation. BAR2 reaches approximately 5.6×10^{-5}; the NN2, NN3, and BAR3 excursions are capped at 2×10^{-3}. The final flux-correction diagnostics reach the 12-pass limit in 23 of 36 cases. Remaining flux can therefore separate the returned state from the unlimited target even when conservation and the relaxed bounds are satisfied. No admissibility repair is activated. Wall-normal momentum at slip-wall vertices can reach 1.3×10^{-4} immediately after LC transfer and is removed by the next flow step. These results concern the specified Riemann configurations and do not establish an LC result for the RANS airfoil cases.